

GAGNOA, Côte d'Ivoire

WMO: 655570

Lat: 6.133N

Lon: 5.950W

Elev: 210

StdP: 98.83

Time Zone: 0.00 (E00)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.7	19.8	11.2	8.5	29.4	14.3	10.4	24.7	4.5	26.1	4.3	26.0	0.1	0	0.255

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	10.0	35.0	24.8	34.2	25.2	33.7	25.3	27.2	32.3	26.9	32.1	26.6	31.6	1.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	21.9	30.3	25.5	21.2	29.5	25.2	20.8	29.1	87.6	32.6	85.9	32.3	84.5	32.0	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.2	3.5	3.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	27.4	28.5	28.3	28.2	27.5	26.5	25.7	25.5	26.6	27.1	27.3	27.0
	DBStd	1.45	1.82	1.37	1.17	1.10	1.19	0.97	0.91	0.89	0.90	0.84	0.74	1.17
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6249	539	518	568	544	544	496	486	479	497	530	521	526
	CDD18.3	3207	281	284	310	294	286	246	228	221	247	272	271	268
	CDH23.3	27558	2780	2940	3123	2982	2627	1817	1469	1348	1758	2067	2269	2377
	CDH26.7	10850	1299	1423	1428	1312	1022	552	360	312	566	723	889	963
(14) Wind	WSAvg	1.2	1.1	1.1	1.1	1.1	1.1	1.2	1.3	1.5	1.4	1.3	1.2	0.8
(15) Precipitation	PrecAvg	1397	21	66	124	181	182	201	88	88	158	160	101	40
	PrecMax	2000	151	158	271	326	355	441	400	273	350	320	234	172
	PrecMin	954	0	2	14	85	79	45	8	11	14	65	13	0
	PrecStd	229	31	38	53	57	62	85	68	55	84	61	59	42
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	36.2	35.6	34.9	34.0	32.2	31.1	31.0	32.2	33.0	33.3	33.9
		MCWB	23.0	23.7	25.9	26.6	27.1	26.2	25.1	24.6	25.2	25.9	25.9	23.4
	2%	DB	34.2	35.2	34.6	34.0	33.0	31.2	30.2	30.1	31.3	32.1	32.5	32.9
		MCWB	23.3	24.4	26.1	26.6	26.7	25.9	24.8	24.6	25.1	25.5	25.8	24.1
	5%	DB	33.5	34.3	33.8	33.2	32.2	30.5	29.6	29.3	30.7	31.4	31.9	32.0
		MCWB	23.5	24.7	26.0	26.4	26.4	25.6	24.6	24.5	25.1	25.5	25.7	24.3
	10%	DB	32.5	33.4	32.9	32.5	31.4	29.7	28.8	28.6	29.8	30.6	31.1	31.2
		MCWB	23.5	24.8	25.9	26.2	26.2	25.4	24.4	24.4	24.9	25.4	25.6	24.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	27.2	27.6	27.8	27.7	27.1	26.2	25.7	26.2	26.6	27.0	26.5
		MCDB	31.8	33.4	33.2	32.8	32.3	30.8	29.8	29.3	30.6	30.9	31.6	30.9
	2%	WB	25.7	26.6	27.0	27.1	27.2	26.4	25.5	25.2	25.7	26.1	26.5	26.0
		MCDB	31.1	32.6	32.8	32.7	31.9	30.2	29.2	28.9	30.1	30.7	31.1	30.7
	5%	WB	25.4	26.1	26.6	26.7	26.7	25.9	25.0	24.8	25.3	25.8	26.1	25.7
		MCDB	30.5	31.9	32.3	32.1	31.3	29.7	28.6	28.4	29.6	30.3	30.8	30.1
	10%	WB	25.0	25.6	26.2	26.4	26.3	25.5	24.6	24.4	25.0	25.5	25.8	25.3
		MCDB	29.9	30.9	31.8	31.5	30.6	29.0	27.9	27.8	29.1	29.8	30.3	29.4
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.0	10.0	9.5	8.5	7.3	6.8	6.7	7.6	8.1	8.8	9.9
		MCDBR	12.1	11.9	11.0	10.3	9.3	8.2	7.8	7.7	8.5	8.9	9.4	10.7
	5% WB	MCWBR	3.5	3.4	3.9	3.7	3.7	3.5	3.1	3.1	3.1	3.2	3.6	3.7
		MCWBR	10.1	10.6	10.4	9.9	9.0	7.7	7.1	7.0	8.0	8.5	9.0	9.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.738	0.720	0.748	0.587	0.507	0.486	0.478	0.461	0.454	0.479	0.544	0.660
		taud	1.586	1.651	1.608	1.944	2.148	2.214	2.238	2.291	2.322	2.243	2.010	1.740
	Ebn at Noon	Ebn at Noon	633	663	650	752	796	801	813	842	860	841	777	679
		Edh at Noon	273	262	275	193	154	141	139	135	133	143	177	231
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.82	4.80	5.12	5.21	4.84	4.08	3.88	3.78	4.12	4.50	4.59	4.49
	RadStd	RadStd	0.23	0.18	0.21	0.21	0.18	0.21	0.19	0.23	0.17	0.15	0.15	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page